YIFEI LIU'S PORTFOLIO CASE STUDY

Bank ALM / IRRBB & Liquidity Stress Testing

A Python, SQL and Excel Portfolio Study of Interest Rate Risk, EVE / NII Sensitivity and 90-Day Liquidity Stress

PROJECT SNAPSHOT

USD 100M Synthetic Banking Book	**USD -3.167M** Worst Delta EVE
USD -32.2K Worst 12M Delta NII	**≥90 Days** Combined Stress Survival

PROJECT INFORMATION

Field	Content
Author	Yifei Liu
Target roles	Treasury / ALM Analyst | Market Risk Analyst | Risk Analytics / Model Validation
Core stack	Python | SQL | DuckDB | Excel
Risk focus	ALM | IRRBB | Liquidity Stress Testing
Portfolio	938 synthetic contract / group records
Positioning	Reproducible ALM / IRRBB and liquidity stress-testing portfolio project

Repository
https://github.com/Liuyifi/bank-alm-irrbb-liquidity-stress-testing-python

Scan the QR code to review the code, evidence and reproducible workflow.

Executive Summary

This portfolio case study assesses earnings sensitivity, long-term economic-value exposure and short-term liquidity resilience for a stylized banking book. I built the deterministic portfolio, contract-level cash-flow engine, scenario framework, daily liquidity ledger, DuckDB reconciliation layer and Excel management pack from reviewed public-source inputs.

Portfolio evidence	Evidence
Business problem	Measure repricing mismatch, 12-month NII sensitivity, full-life EVE exposure and 90-day liquidity resilience together.
What I built	A deterministic 938-record balance sheet, cash-flow engine, seven rate scenarios, three liquidity stresses and a reconciliation framework.
Key IRRBB result	Parallel up 200 bps is the dominant tested economic-value shock: Delta EVE is USD -3.167m.
Earnings result	Worst 12-month Delta NII is much smaller at USD -32.2k under the default behavioral assumptions.
Liquidity result	All scenarios survive at least 90 days; Combined finishes with USD 1.017m available cash.
Validation	64 / 64 pytest; 52 / 52 automated controls; 17 / 17 metric reconciliations; Python / SQL difference below USD 0.01.

Main interpretation. The book is materially more exposed through long-term economic value than through first-year earnings under the default assumptions.

Main limitation. Behavioral deposit beta, maturity and runoff are modeled assumptions rather than empirically estimated bank parameters, and they materially affect both NII and EVE.

Professional boundary. This is a synthetic portfolio study, not a production bank model or a regulatory compliance calculation.

Business Problem, Data and Portfolio Construction

Banks monitor repricing, NII, EVE and liquidity together because short-horizon earnings and long-horizon value can move differently, while funding stress can constrain the actions available to manage either exposure. This project links those views in one reproducible study.

Area	Portfolio design
Public calibration	Federal Reserve H.8, release 2026-10-16; observation 2026-10-07; all US commercial banks, not seasonally adjusted.
Market rates	US Treasury Daily Treasury Par Yield Curve Rates, observation 2026-10-22.
Portfolio size	USD 100m of synthetic assets.
Record count	938 synthetic contract / group records generated with deterministic seed 20261023.
Products	Loans, securities, deposits, borrowings and residual balance-sheet categories.
Behavioral assumptions	Deposit beta, NMD maturity, runoff, liquidity haircuts, funding nonrenewal and commitment draws.

OBSERVED AND MODELED

Observed public evidence	Modeled portfolio assumptions
Treasury par yields; H.8 cash, securities, gross loans, deposits, borrowings and residual balance-sheet shares.	Product splits, contractual dates, coupons, reset schedules, amortization, deposit behavior, runoff, haircuts and commitment draws.

Governance distinction. Public aggregates anchor the size and broad composition of the book; they do not reveal a bank-specific contractual portfolio. All position-level details remain synthetic and traceable to documented assumptions.

ANALYTICAL WORKFLOW



Repricing Gap and Balance-Sheet Structure

Rate-sensitive principal is assigned to the first contractual reset, or final maturity for fixed-rate instruments. Non-maturity deposits reprice monthly and therefore concentrate in the shortest bucket; non-rate-sensitive balances are excluded from RSA and RSL.

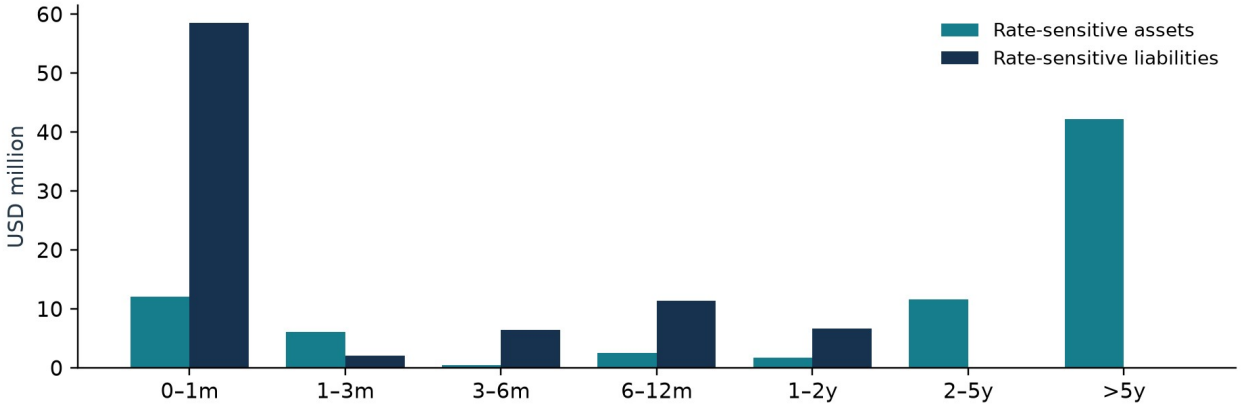


Figure 1. Rate-sensitive assets and liabilities by next repricing bucket.

Bucket	RSA USD m	RSL USD m	Gap USD m
Overnight to 1 month	12.14	58.55	-46.42
1 to 3 months	6.15	2.06	+4.09
3 to 6 months	0.44	6.45	-6.01
6 to 12 months	2.49	11.34	-8.85
1 to 2 years	1.73	6.62	-4.89
2 to 5 years	11.60	0.00	+11.60
More than 5 years	42.15	0.00	+42.15

- The overnight-to-one-month gap is USD -46.42m: more funding than assets can reprice immediately.
- Assets beyond five years total approximately USD 42.15m with no rate-sensitive liabilities in that bucket, creating long-duration value exposure.
- Gap direction alone does not determine NII because deposit pass-through depends on the modeled NMD beta.

12-Month NII Sensitivity

The 12-month analysis follows the existing book as it runs off. Fixed coupons remain unchanged until maturity; floating-rate assets reset on schedule; matured principal is not reinvested. The result is an existing-book sensitivity, not a forecast for a growing balance sheet.

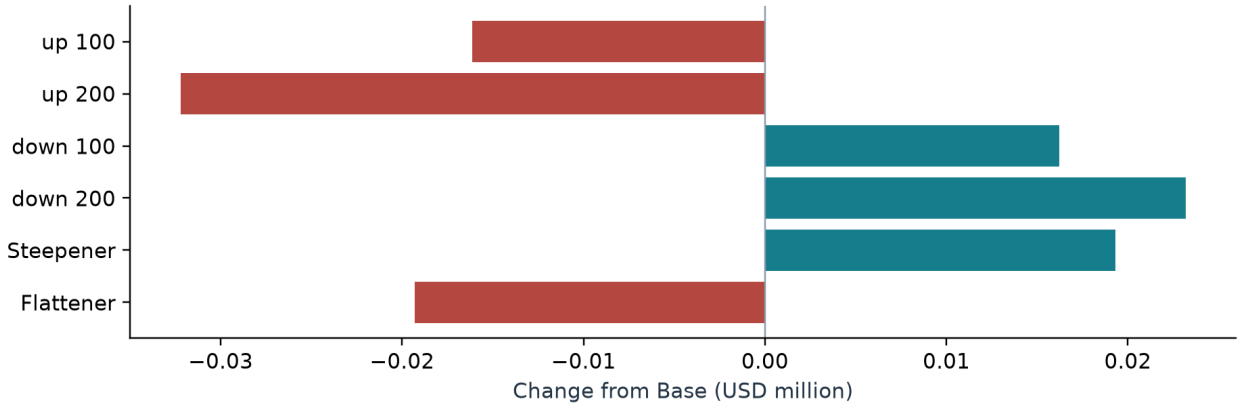


Figure 2. Twelve-month Delta NII under internal illustrative scenarios.

Scenario	NII USD m	Delta USD k	Delta / Base
Base	2.568	+0.0	0.00%
Parallel up 100 bps	2.551	-16.1	-0.63%
Parallel up 200 bps	2.535	-32.2	-1.25%
Parallel down 100 bps	2.584	+16.2	0.63%
Parallel down 200 bps	2.591	+23.2	0.90%
Steepener	2.587	+19.3	0.75%
Flattener	2.548	-19.3	-0.75%

Key result. Parallel up 200 bps produces the worst tested 12-month change at USD -32.2k, small relative to EVE sensitivity.

Why this matters. A small default earnings impact does not mean the book is hedged. NMD beta sensitivity can reverse the NII direction because modeled deposit pass-through changes the funding-cost response.

EVE / IRRBB Sensitivity

EVE measures the discounted value of asset cash flows less liability cash flows over the full modeled life. Floating coupons are reprojected under each stressed curve and NMD principal follows its behavioral maturity schedule; equity is excluded as a liability.

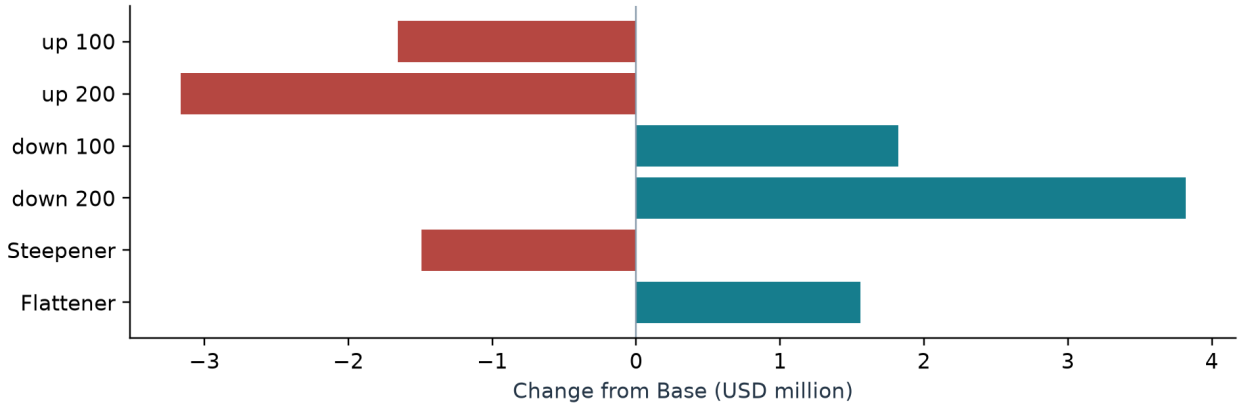


Figure 3. Delta EVE under internal illustrative scenarios.

Scenario	EVE USD m	Delta USD m	Delta / equity proxy
Base	12.927	+0.000	0.00%
Parallel up 100 bps	11.270	-1.658	-17.60%
Parallel up 200 bps	9.760	-3.167	-33.62%
Parallel down 100 bps	14.749	+1.822	19.34%
Parallel down 200 bps	16.744	+3.817	40.52%
Steepener	11.437	-1.491	-15.82%
Flattener	14.484	+1.557	16.52%

Worst tested scenario. Parallel up 200 bps reduces EVE by USD -3.167m, equal to -33.62% of the stylized equity residual proxy.

Interpretation. Long-lived fixed-rate asset cash flows remain exposed to higher discount rates even when first-year coupon income changes only modestly.

Boundary. The equity residual proxy is not regulatory capital, and this result is not a BCBS supervisory outlier test or a production IRRBB regulatory calculation.

90-Day Liquidity Stress Testing

A daily ledger starts with cash and recognizes stress outflows, contractual inflows and timed government-security monetization. Securities can be monetized once after haircuts; later cash flows are excluded to prevent double counting.

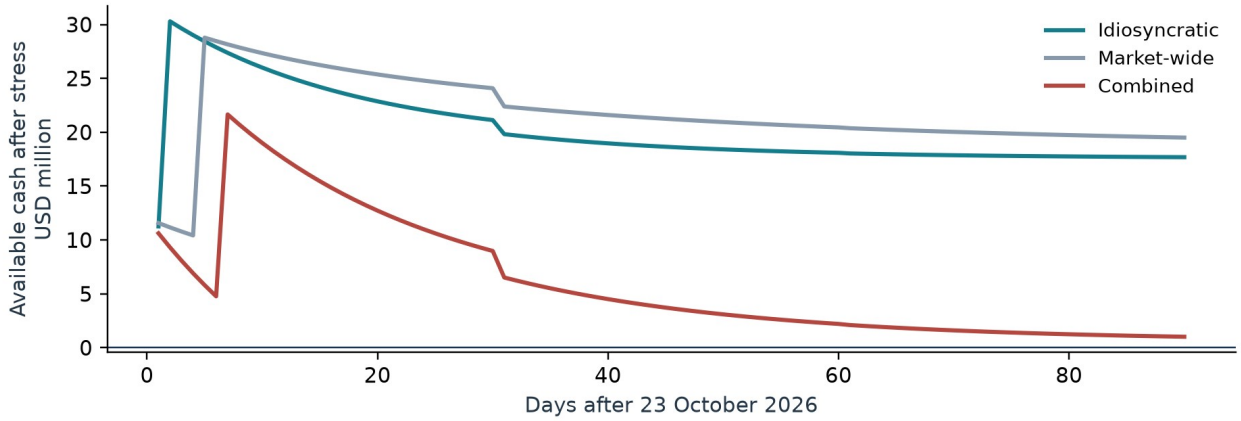


Figure 4. Available cash through the 90-day stress horizon.

Scenario	Outflows USD m	Recognized inflows USD m	End buffer USD m	Survival
Idiosyncratic	15.13	1.07	17.686	≥90 days
Market-wide	12.22	1.02	19.505	≥90 days
Combined	29.92	1.07	1.017	≥90 days

Key result. All three scenarios survive at least 90 days. Combined finishes with only USD 1.017m of available cash, so the result is right-censored rather than evidence of indefinite survival.

Stress mechanics. The scenarios combine retail runoff, commitment drawdown, term nonrenewal, wholesale funding loss, security haircuts, monetization timing and an internal daily 75% inflow cap.

Implementation takeaway. The implementation uses a daily liquidity stock-and-flow ledger rather than only static scenario totals.

Validation and Control Framework

The validation design checks source integrity, calculation consistency, independent control logic where feasible, saved Office artifacts and current-state metric reconciliation. This separates generated outputs from evidence that the outputs are internally consistent.

64 / 64 pytest	**52 / 52** Automated Acceptance Controls	**30 / 30** Core Engine / Data Controls
17 / 17 Current Metric Reconciliations	**USD 0.0000008158** Maximum Python / SQL Difference	

SQL CONTROL STRUCTURE

Classification	Controls	Meaning
Independent recomputation	Balance sheet; repricing gap	SQL recomputes results from position-level inputs without Python result columns.
Partial recomputation	NII; EVE; liquidity	SQL independently aggregates or reconstructs controls but reuses projected cash flows or discount inputs.

ARTIFACT AND GOVERNANCE EVIDENCE

Artifact	Evidence
Excel management pack	12 sheets; 15 native tables; 4 native charts; 88 cached formulas; no external links or formula errors.
Portfolio report	Nine-page render; reconciled core metrics; source names; repository and QR; author identity; model-boundary language; no local paths or unfinished text.
Source governance	Versioned source lock, raw-to-processed lineage and fail-closed validation before analysis or artifact writes.

Validation boundary. The controls establish implementation consistency under documented synthetic assumptions; they do not empirically validate deposit behavior or turn the SQL layer into a second independent valuation engine.

Skills Demonstrated and Responsible Interpretation

SKILLS DEMONSTRATED

Capability	Evidence demonstrated in this project
Risk analytics	NII sensitivity; EVE / IRRBB analysis; repricing gap; 90-day liquidity stress; NMD behavioral sensitivity.
Programming and data	Python cash-flow modelling; SQL / DuckDB reconciliation; reproducible pipeline; structured controls; source lineage and fail-closed validation.
Risk communication & reporting	Excel management pack; risk visualization; executive interpretation; concise model-boundary documentation.

KEY LIMITATIONS

- The banking book is synthetic and calibrated to public industry aggregates rather than bank-level records.
- Deposit beta, behavioral maturity and runoff are modeled assumptions, not empirically estimated parameters.
- Mortgage prepayment, embedded options and dynamic customer migration are not modeled.
- Derivative hedges, hedge costs and optimization are outside scope.
- Liquidity results are internal scenario proxies; no formal LCR or NSFR is calculated.
- The study is not a BCBS standardized IRRBB implementation or a regulatory capital assessment.

REPOSITORY AND SUPPORTING EVIDENCE

Full methodology, source lineage, validation evidence, reviewed outputs and reproducible code are available in the public repository.

https://github.com/Liuyifi/bank-alm-irrbb-liquidity-stress-testing-python

This report is designed as a standalone recruitment attachment; the repository provides the deeper technical and audit trail for interview discussion.